

Physical Chemistry (I)

Lecture 14

Statistical Mechanics
Phase



In the Last Class

- Partition function
 - Definition
 - Partition function and thermodynamics
- Classical statistical mechanics
 - Phase space
 - Calculation of partition function

Partition Function

In general, for a microstate s with energy E ,

$$P(s) \propto e^{-E(s)/k_{\text{B}}T}$$

or, using a constant q ,

$$P(s) = \frac{1}{q} e^{-E(s)/k_{\text{B}}T}$$

$$q = \sum_s e^{-E(s)/k_{\text{B}}T}$$

(Partition function)

Partition Function and ...

- Helmholtz energy: $A = -k_{\text{B}}T \ln q$
- Internal energy: $U = -\partial(\ln q)/\partial\beta$
- Entropy: $S = (U - A)/T$

“A partition function contains
all the **thermodynamic information** about a system.”

Last class

Derived Functions

We know Helmholtz energy from the partition function,

$$A = -k_{\text{B}}T \ln q$$

Using $dA = -SdT - pdV$,

$$S = -\left(\frac{\partial A}{\partial T}\right)_V, \quad p = -\left(\frac{\partial A}{\partial V}\right)_T$$



Phase Space

Space of **all possible microstates** of a system

- Classical mechanics:

Space of all possible values of
position and **momentum** variables

Last class

Classical Partition Function

$$q = \sum_s e^{-E(s)/k_B T} \rightarrow q = \frac{1}{h} \int \int e^{-E(x,p)/k_B T} dx dp$$

for a 1-dimensional single particle

Last class

Calculation of Q

For indistinguishable particles in 3D space,

$$Q = \frac{1}{N! h^{3N}} \left[\int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p} \right]^N = \frac{1}{N!} q^N$$

where

$$q = \frac{1}{h^3} \int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p}$$

Perfect Gas

- No interactions (no potential energy)
- Total energy = kinetic energy ($p^2/2m$)
- Bound in a box of volume V
 - Range of $x = (0, a)$
 - Range of $y = (0, b)$
 - Range of $z = (0, c)$
 - $abc = V$
- Range of momentum = $(-\infty, \infty)$

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q of Perfect Gas

$$q = \frac{1}{h^3} \int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p}$$

Since $E(\vec{r}, \vec{p}) = p^2/2m$,

$$q = \frac{1}{h^3} \int \int e^{-p^2/2mk_B T} d\vec{r} d\vec{p}$$

$$q = \frac{1}{h^3} \left[\int 1 d\vec{r} \right] \left[\int e^{-p^2/2mk_B T} d\vec{p} \right]$$

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Position Integral

$$q = \frac{1}{h^3} \left[\int 1 \, d\vec{r} \right] \left[\int e^{-p^2/2mk_{\text{B}}T} \, d\vec{p} \right]$$

$$\int 1 \, d\vec{r} = \int_0^c \int_0^b \int_0^a 1 \, dx \, dy \, dz = abc = V$$

Momentum Integral

$$q = \frac{V}{h^3} \left[\int e^{-p^2/2mk_B T} d\vec{p} \right]$$

$$\int e^{-p^2/2mk_B T} d\vec{p} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-p^2/2mk_B T} dp_x dp_y dp_z$$

Since $e^{-p^2/2mk_B T} = e^{-p_x^2/2mk_B T} e^{-p_y^2/2mk_B T} e^{-p_z^2/2mk_B T}$,

$$\left[\int_{-\infty}^{\infty} e^{-p_x^2/2mk_B T} dp_x \right] \left[\int_{-\infty}^{\infty} e^{-p_y^2/2mk_B T} dp_y \right] \left[\int_{-\infty}^{\infty} e^{-p_z^2/2mk_B T} dp_z \right]$$

Gaussian Integral Again

$$q = \frac{V}{h^3} \left[\int_{-\infty}^{\infty} e^{-p_x^2/2mk_B T} dp_x \right]^3$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\pi/\alpha},$

$$q = \frac{V}{h^3} \left[\sqrt{2\pi mk_B T} \right]^3 = V \left(\frac{2\pi mk_B T}{h^2} \right)^{3/2}$$

Q of Perfect Gas

$$q = V \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2}$$

For indistinguishable gas particles,

$$Q = \frac{1}{N!} q^N = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Helmholtz Energy

$$Q = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Now, we can calculate the Helmholtz energy:

$$A = -k_B T \ln Q = -k_B T \ln \left\{ \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2} \right\}$$

$$A = -Nk_B T \ln V + k_B T \ln N! - \frac{3}{2} Nk_B T \ln \left(\frac{2\pi m k_B T}{h^2} \right)$$

Internal Energy

$$Q = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Also the internal energy:

$$U = -\frac{\partial(\ln Q)}{\partial \beta} = -\frac{\partial}{\partial \beta} \ln \left\{ \frac{V^N}{N!} \left(\frac{2\pi m}{\beta h^2} \right)^{3N/2} \right\}$$

$$U = \frac{\partial}{\partial \beta} \ln \beta^{3N/2} + \frac{\partial}{\partial \beta} (\text{terms independent to } \beta) = \frac{3}{2} N k_B T$$

Pressure

$$A = -Nk_B T \ln V + k_B T \ln N! - \frac{3}{2} Nk_B T \ln \left(\frac{2\pi m k_B T}{h^2} \right)$$

Using $p = -(\partial A / \partial V)_T$,

$$p = Nk_B T \frac{\partial(\ln V)}{\partial V} + \frac{\partial}{\partial V} (\text{terms independent to } V)$$

$$p = \frac{Nk_B T}{V}$$

One can obtain **entropy** similarly

Real Gas

Particles are no more independent to each other

$$Q \neq q^N / N!$$

$$Q = \frac{1}{N! h^{3N}} \int \cdots \int e^{-E_{\text{tot}}/k_B T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

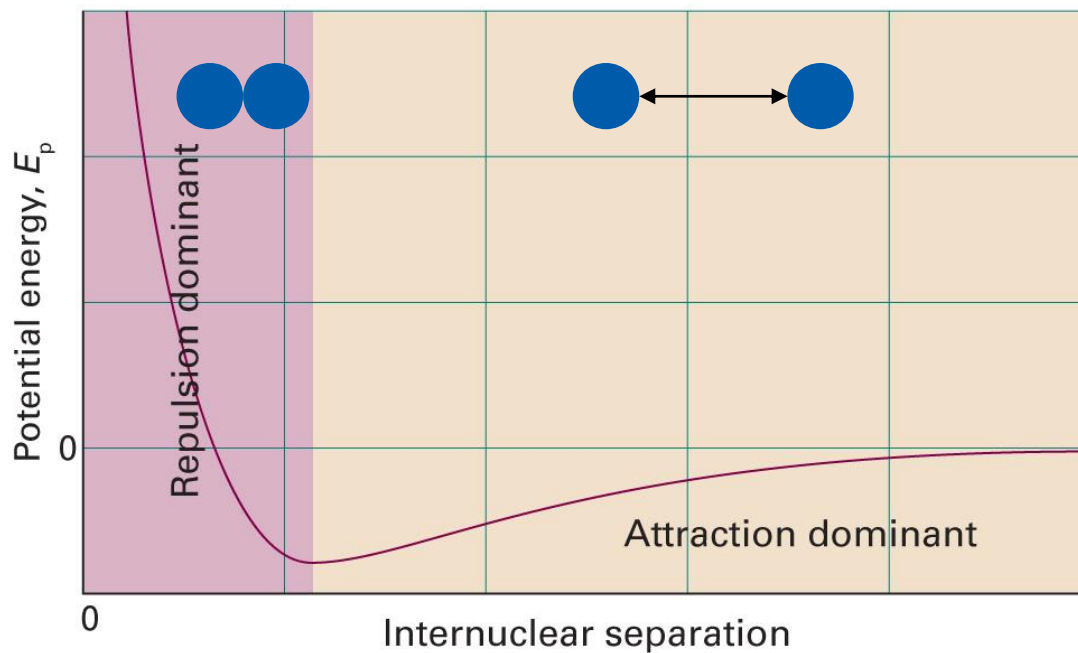
where

$$E_{\text{tot}} = \sum_i \frac{p_i^2}{2m} + U(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N)$$

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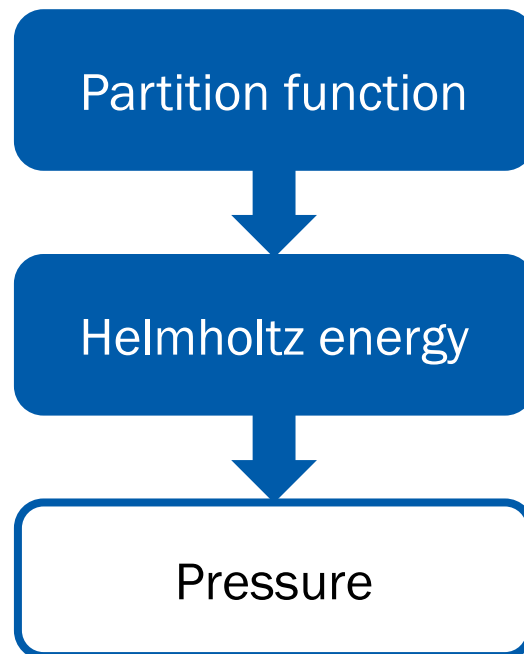
Potential Energy

Pairwise interactions (attraction + repulsion)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1C.1)

Pressure of Real Gas



$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

Equipartition Theorem

For a sample at thermal equilibrium,
the average value of each **quadratic contribution**
to the internal energy is $(1/2)k_B T$.

(Quadratic contribution: e.g., $E \propto p_x^2$ or $E \propto x^2$)

Can we prove this theorem?

Setting up the Stage

$$Q = \frac{1}{N! h^{3N}} \int \cdots \int e^{-E_{\text{tot}}/k_{\text{B}}T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

Assume that

$$E_{\text{tot}} = \sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2$$
$$(A_i, B_j > 0)$$

and that the ranges of both positions and momenta are $(-\infty, \infty)$

Summation to Product

$$Q = \frac{1}{N! h^{3N}} \int \cdots \int e^{-\left(\sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2\right)/k_B T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

$$= \frac{1}{N! h^{3N}} \int \cdots \int \prod_{i=1}^{f_r} e^{-A_i r_i^2/k_B T} \prod_{j=1}^{f_p} e^{-B_j p_j^2/k_B T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

Thus,

$$Q = \frac{1}{N! h^{3N}} \prod_{i=1}^{f_r} \int_{-\infty}^{\infty} e^{-A_i r_i^2/k_B T} dr_i \prod_{j=1}^{f_p} \int_{-\infty}^{\infty} e^{-B_j p_j^2/k_B T} dp_j$$

Gaussian Integrals

$$Q = \frac{1}{N! h^{3N}} \prod_{i=1}^{f_r} \int_{-\infty}^{\infty} e^{-A_i r_i^2 / k_B T} dr_i \prod_{j=1}^{f_p} \int_{-\infty}^{\infty} e^{-B_j p_j^2 / k_B T} dp_j$$

Using

$$\int_{-\infty}^{\infty} e^{-A_i r_i^2 / k_B T} dr_i = \sqrt{\frac{\pi k_B T}{A_i}}, \quad \int_{-\infty}^{\infty} e^{-B_j p_j^2 / k_B T} dp_j = \sqrt{\frac{\pi k_B T}{B_j}},$$

we obtain

$$Q = \frac{1}{N! h^{3N}} \prod_{i=1}^{f_r} \left(\frac{\pi k_B T}{A_i} \right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{\pi k_B T}{B_j} \right)^{1/2}$$

In Terms of β

$$Q = \frac{1}{N! h^{3N}} \prod_{i=1}^{f_r} \left(\frac{\pi k_B T}{A_i} \right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{\pi k_B T}{B_j} \right)^{1/2}$$

We can write the partition function as follows:

$$Q = \frac{\pi^{(f_r+f_p)/2}}{N! h^{3N}} \left(\frac{1}{\beta} \right)^{(f_r+f_p)/2} \prod_{i=1}^{f_r} \left(\frac{1}{A_i} \right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{1}{B_j} \right)^{1/2}$$

Internal Energy

$$Q = \frac{\pi^{(f_r+f_p)/2}}{N! h^{3N}} \left(\frac{1}{\beta}\right)^{(f_r+f_p)/2} \prod_{i=1}^{f_r} \left(\frac{1}{A_i}\right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{1}{B_j}\right)^{1/2}$$

The internal energy can be calculated from

$$U = - \left(\frac{\partial \ln Q}{\partial \beta} \right)$$

Now,

$$\ln Q = - \frac{f_r + f_p}{2} \ln \beta + (\text{terms independent to } \beta)$$

Internal Energy

$$\ln Q = -\frac{f_r + f_p}{2} \ln \beta + (\text{terms independent to } \beta)$$

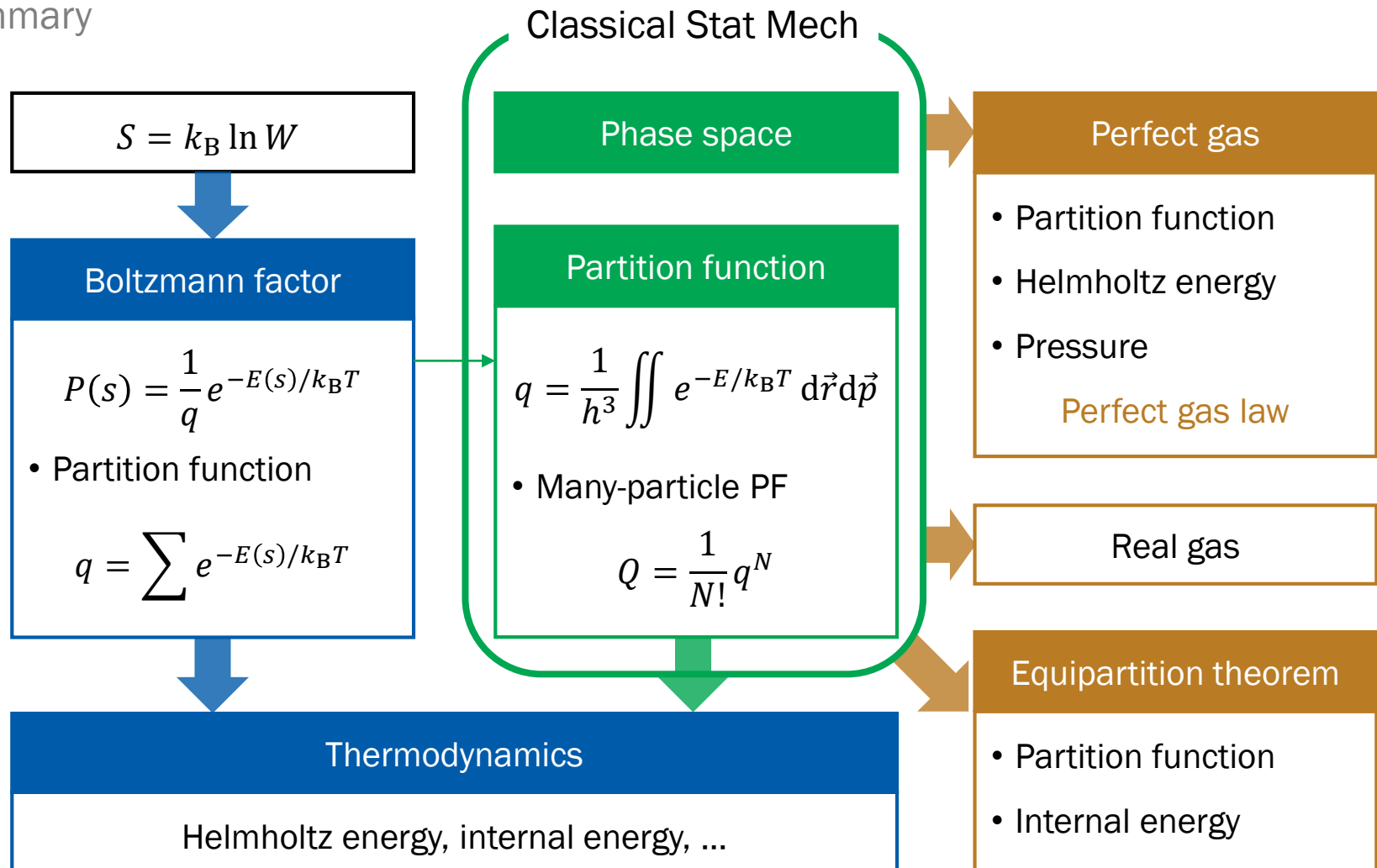
$$U = -\left(\frac{\partial \ln Q}{\partial \beta}\right) = \frac{f_r + f_p}{2} \frac{\partial \ln \beta}{\partial \beta} = \frac{f_r + f_p}{2} \frac{1}{\beta} = \frac{f_r + f_p}{2} k_B T$$

Therefore,

$$U = (f_r + f_p) \frac{k_B T}{2}$$

“The average value of each **quadratic contribution** to the internal energy is $(1/2)k_B T$.”

Summary



7. Phase Transitions

Chapter Overview

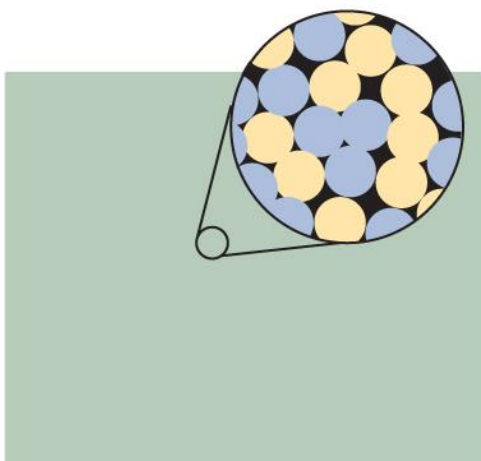
- Phase and chemical potential
 - Phase and phase transition
 - Chemical potential
 - Phase diagram and phase rule
- van der Waals fluid
 - Liquid-gas phase transition
 - Supercritical fluid
 - Critical point and the principle of corresponding states
- Thermodynamic aspects of phase transitions



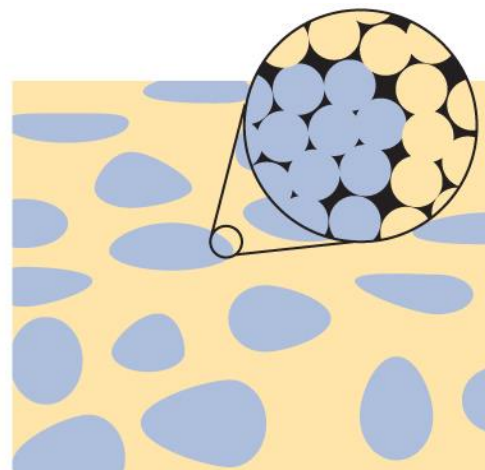
Phase

A form of matter that is **uniform** throughout
in **chemical composition** and **physical state**

Solution: Single Phase



solution
(one phase)



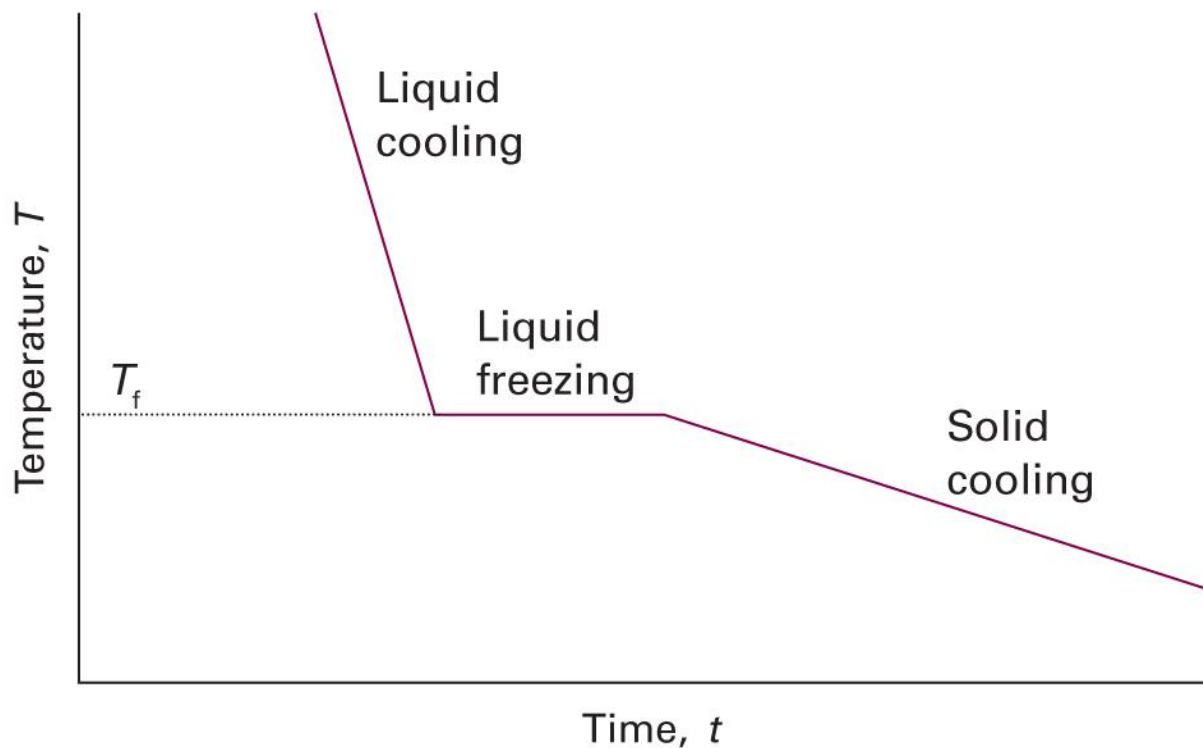
dispersion
(two phases)

Phase Transition

Spontaneous conversion
of one phase into another phase

- occurs at transition temperature T_{trs} (for given p)
- At T_{trs} , the two phases are at equilibrium

Thermal Analysis



Two Phases at Equilibrium

$$dU = TdS - pdV$$

- Mechanistic equilibrium: $p_1 = p_2$
 - Conjugate extensive variable: dV
- Thermal equilibrium: $T_1 = T_2$
 - Conjugate extensive variable: $dS = \delta q/T$
- Diffusive equilibrium: ?
 - Conjugate extensive variable: dn

Chemical Potential

- Diffusive equilibrium: $\mu_1 = \mu_2$
 - Conjugate extensive variable: dn
- (Modified) differentials of thermodynamic potentials

$$dU = TdS - pdV + \mu dn$$

$$dH = TdS + Vdp + \mu dn$$

$$dA = -SdT - pdV + \mu dn$$

$$dG = -SdT + Vdp + \mu dn$$

3 Independent Variables

$$dU = TdS - pdV + \mu dn$$

This implies that $U = U(S, V, n)$, which leads to

$$dU = \left(\frac{\partial U}{\partial S}\right)_{V,n} dS + \left(\frac{\partial U}{\partial V}\right)_{S,n} dV + \left(\frac{\partial U}{\partial n}\right)_{S,V} dn$$

Comparison gives

$$\left(\frac{\partial U}{\partial S}\right)_{V,n} = T, \quad \left(\frac{\partial U}{\partial V}\right)_{S,n} = -p, \quad \left(\frac{\partial U}{\partial n}\right)_{S,V} = \mu$$

Meaning of μ

$$dG = -SdT + Vdp + \mu dn$$

This implies that $G = G(T, p, n)$.

G is extensive, and n is the only extensive variable. Thus,

$$G(T, p, n) = ng(T, p)$$

and

$$\left(\frac{\partial G}{\partial n}\right)_{T,p} = \left[\frac{\partial\{ng(T, p)\}}{\partial n}\right]_{T,p} = g(T, p)$$

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Meaning of μ

$$g(T, p) = \frac{G}{n} = \left(\frac{\partial G}{\partial n} \right)_{T, p}$$

However,

$$dG = -SdT + Vdp + \mu dn$$

gives

$$\mu = \left(\frac{\partial G}{\partial n} \right)_{T, p} = g(T, p) = \frac{G}{n}$$

(molar Gibbs energy)

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Thermodynamic Potentials

$$\mu = G/n$$

can be written as

$$G = \mu n$$

From this equation, we can obtain

$$H = G + TS = \mu n + TS$$

$$A = G - pV = \mu n - pV$$

$$U = G + TS - pV = \mu n + TS - pV$$



Gibbs-Duhem Equation

$$U = \mu n + TS - pV$$

Differentiating both sides,

$$dU = \mu dn + n d\mu + T dS + S dT - p dV - V dp$$

but we know that

$$dU = T dS - p dV + \mu dn$$

Thus,

$$S dT - V dp + n d\mu = 0 \text{ (Gibbs-Duhem equation)}$$

$$d\mu = -S_m dT + V_m dp$$



Two Systems in Contact

$$dG = -SdT + Vdp + \mu dn$$

At constant T and p ,

$$dG = \mu dn$$

For two systems in contact,

$$dG_1 = \mu_1 dn_1$$

$$dG_2 = \mu_2 dn_2$$

The total change in Gibbs energy is

$$dG = dG_1 + dG_2 = \mu_1 dn_1 + \mu_2 dn_2$$

Diffusive Equilibrium

$$dG = dG_1 + dG_2 = \mu_1 dn_1 + \mu_2 dn_2$$

If the two subsystems consist of the same type of particles and can exchange particles,

$$dn_1 + dn_2 = 0$$

Hence,

$$dG = \mu_1 dn_1 - \mu_2 dn_1 = (\mu_1 - \mu_2) dn_1$$

At equilibrium, $dG = 0$, and

$$\mu_1 = \mu_2$$

What If $\mu_1 \neq \mu_2$?

$$dG = \mu_1 dn_1 - \mu_2 dn_1 = (\mu_1 - \mu_2) dn_1$$

- If $\mu_1 > \mu_2$, for $dG < 0$ (spontaneous change),

$$dn_1 < 0 \text{ and } dn_2 = -dn_1 > 0,$$

which means that the particles move from system 1 to 2

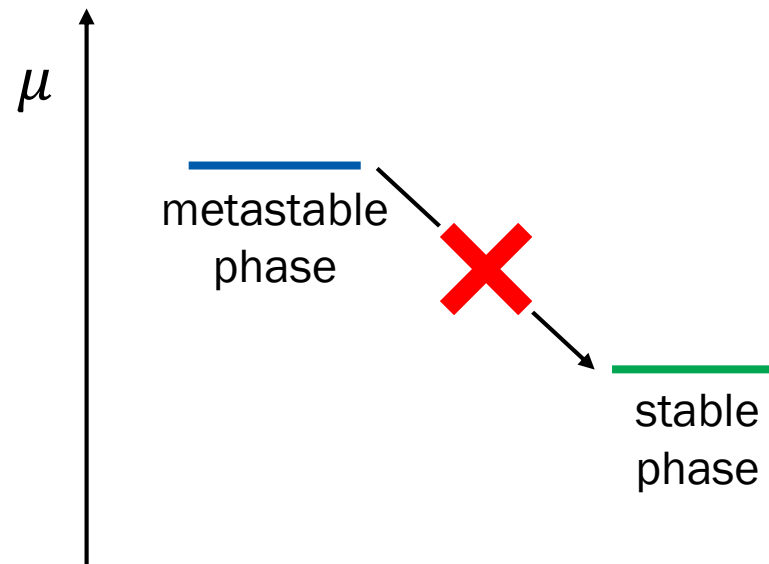
- If $\mu_1 < \mu_2$, for $dG < 0$ (spontaneous change),

$$dn_1 > 0 \text{ and } dn_2 = -dn_1 < 0,$$

which means that the particles move from system 2 to 1

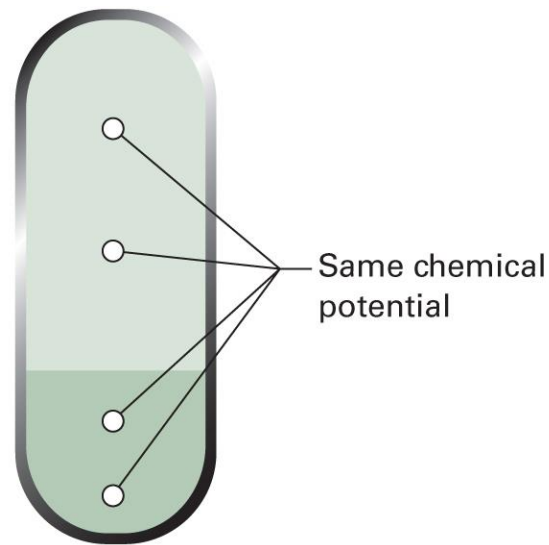
Metastable Phase

Thermodynamically unstable phases that persist because the transition is kinetically hindered



Criteria for Phase Stability

At equilibrium, the **chemical potential** of a substance is the **same** in and throughout every phase in the system

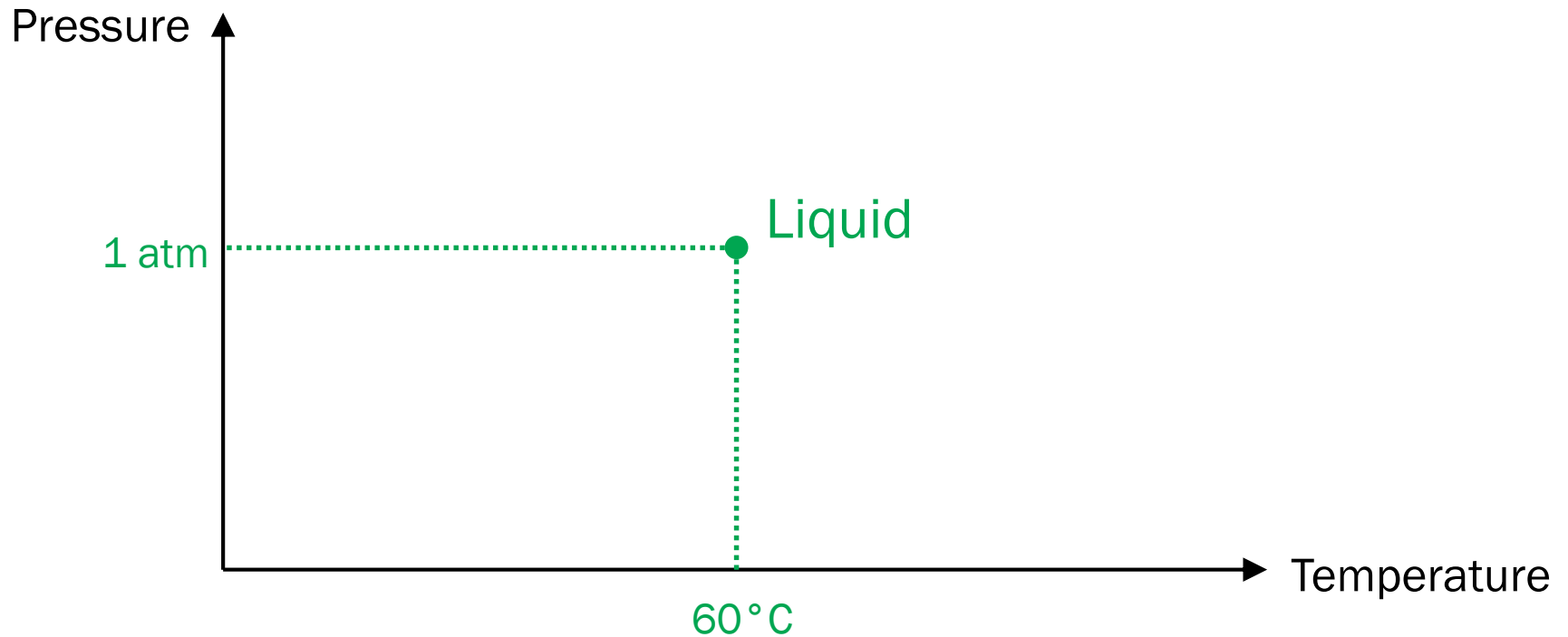


Phase Diagram

A diagram that shows the regions at which its various phases are thermodynamically stable

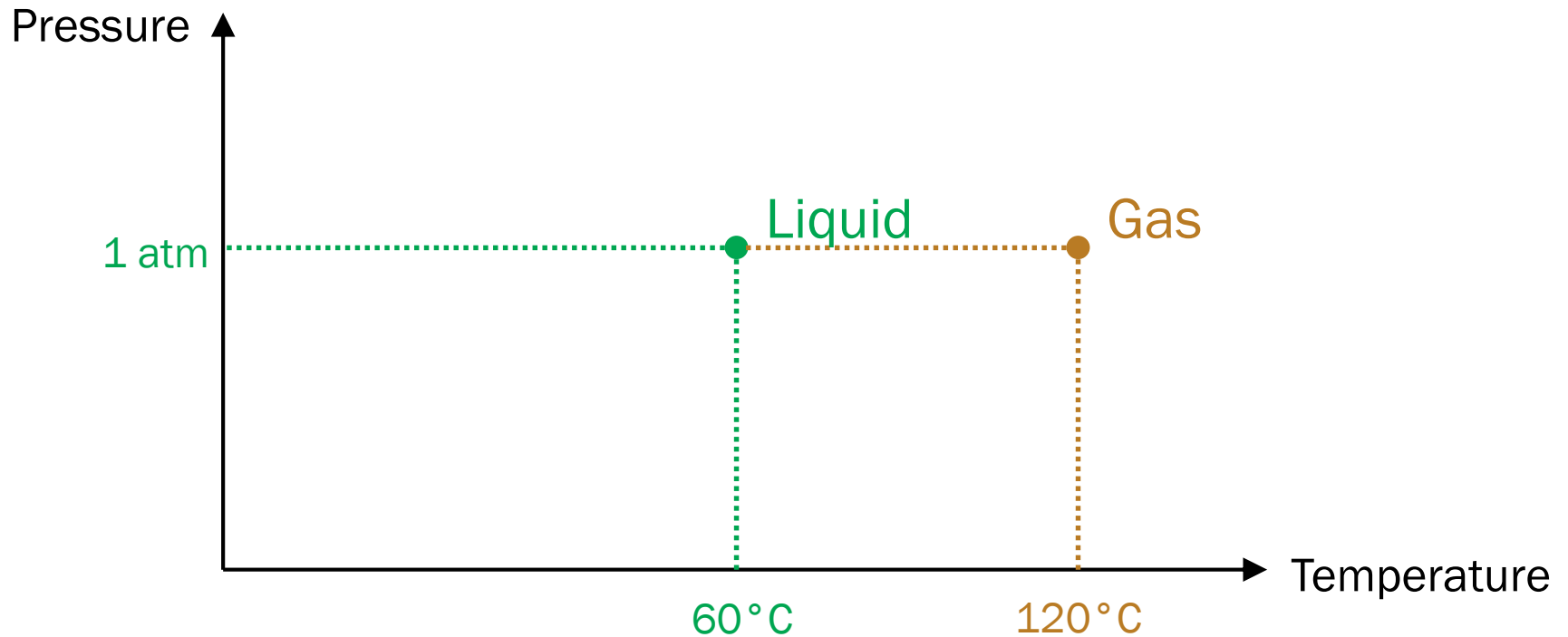
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Drawing Phase Diagram



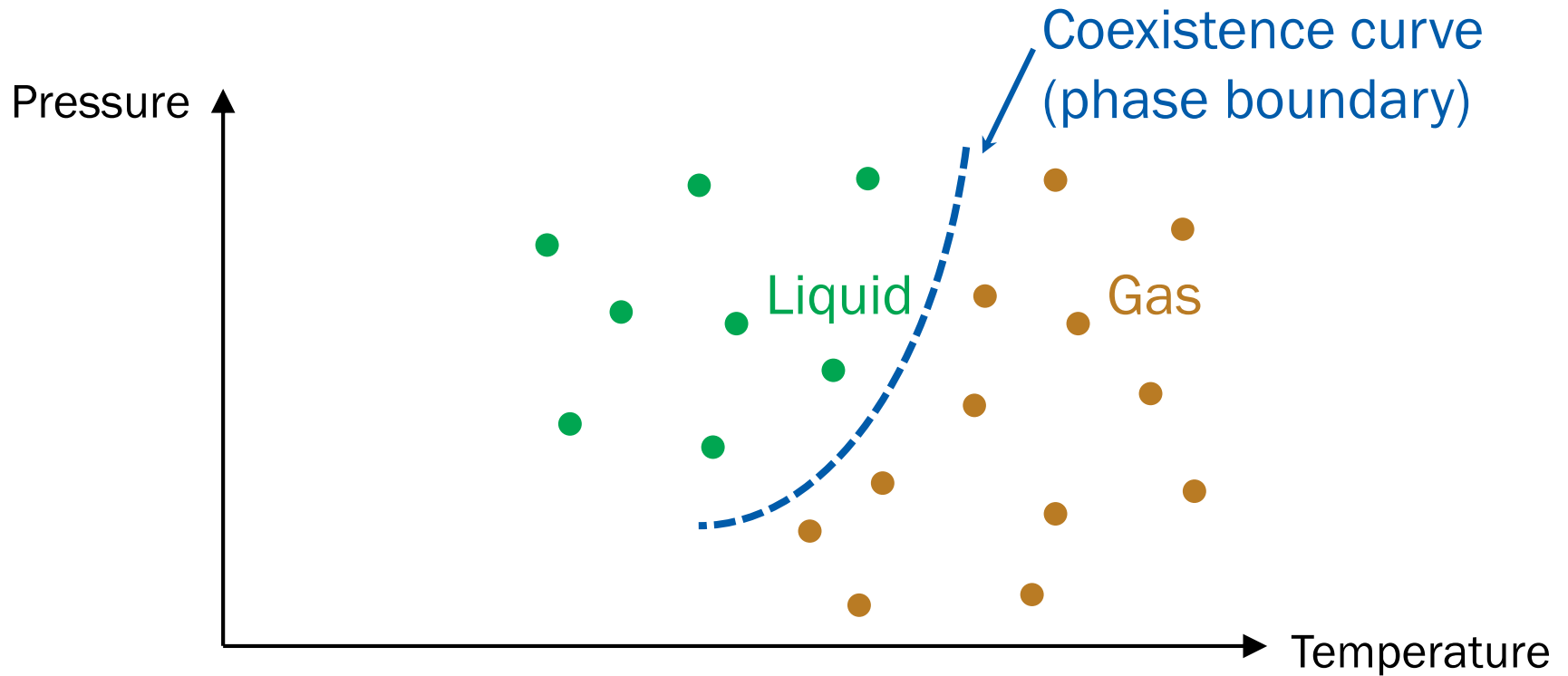
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Drawing Phase Diagram

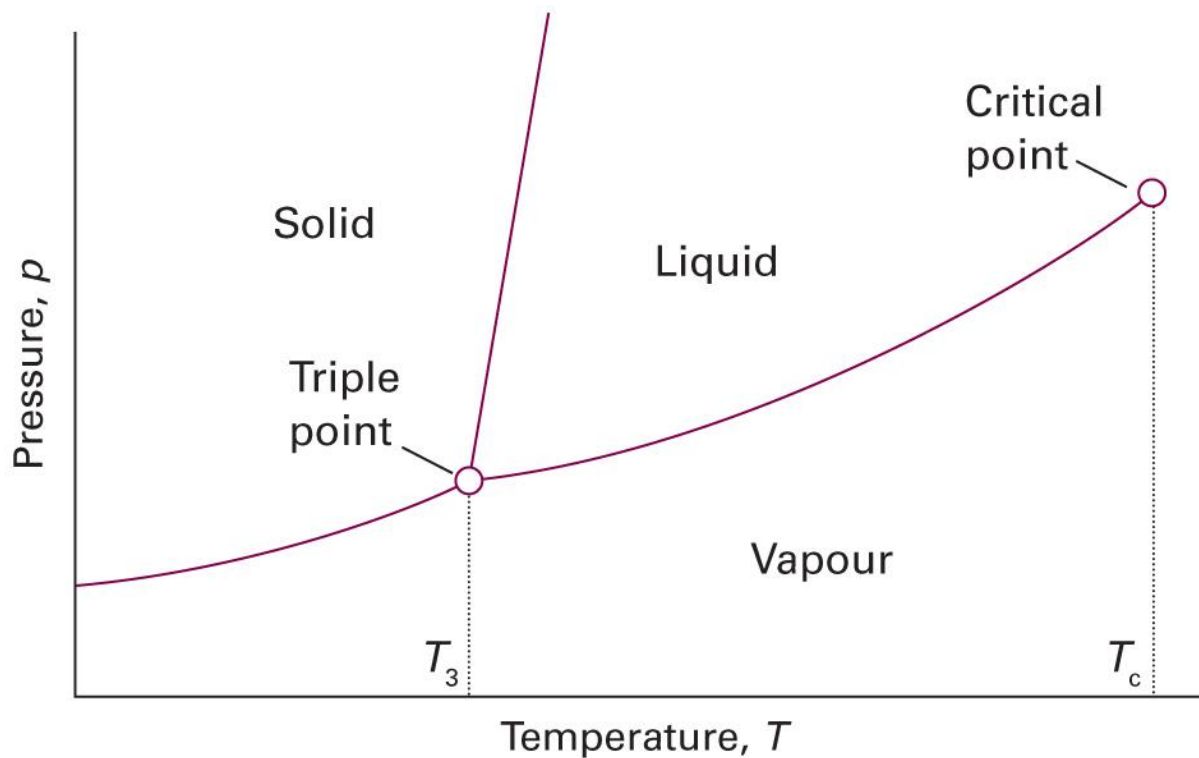


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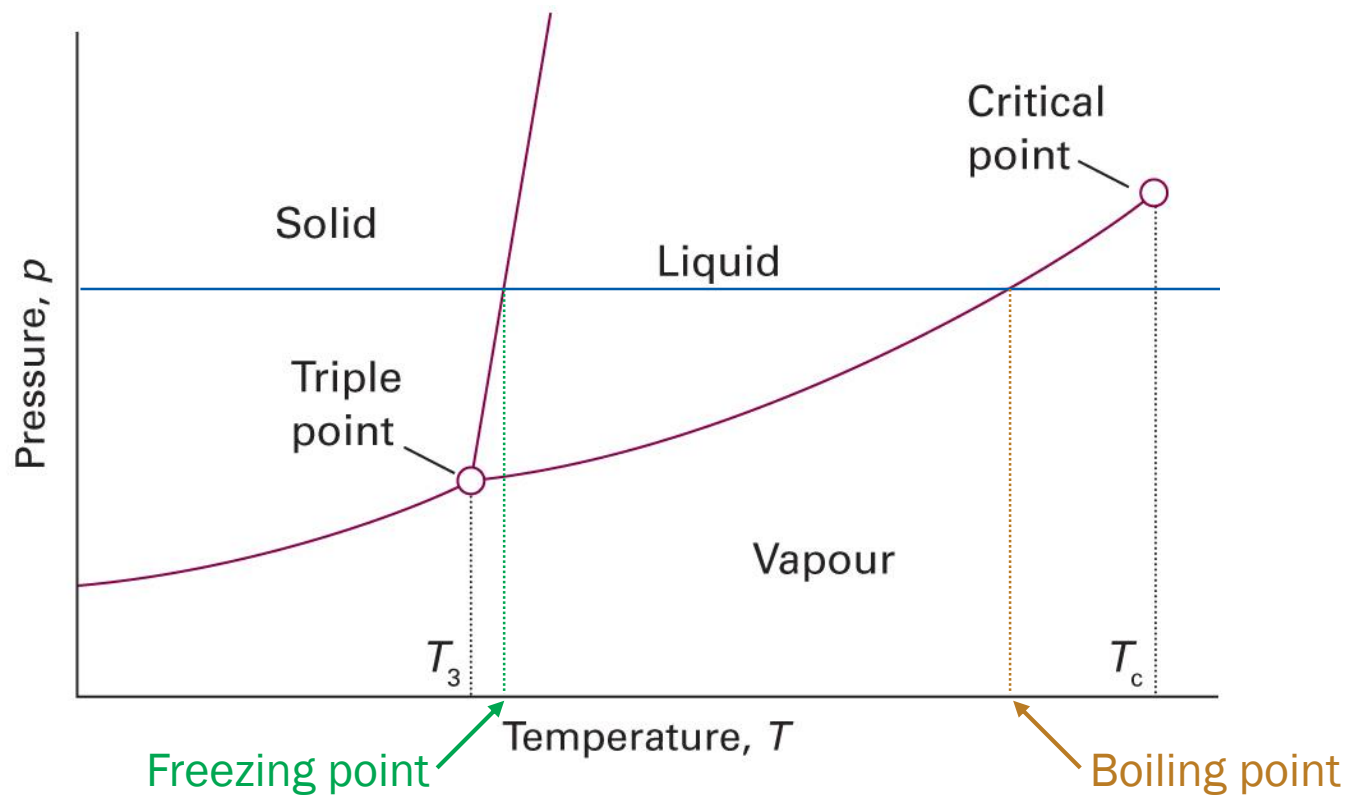
Drawing Phase Diagram



Typical Phase Diagram



Freezing and Boiling Points



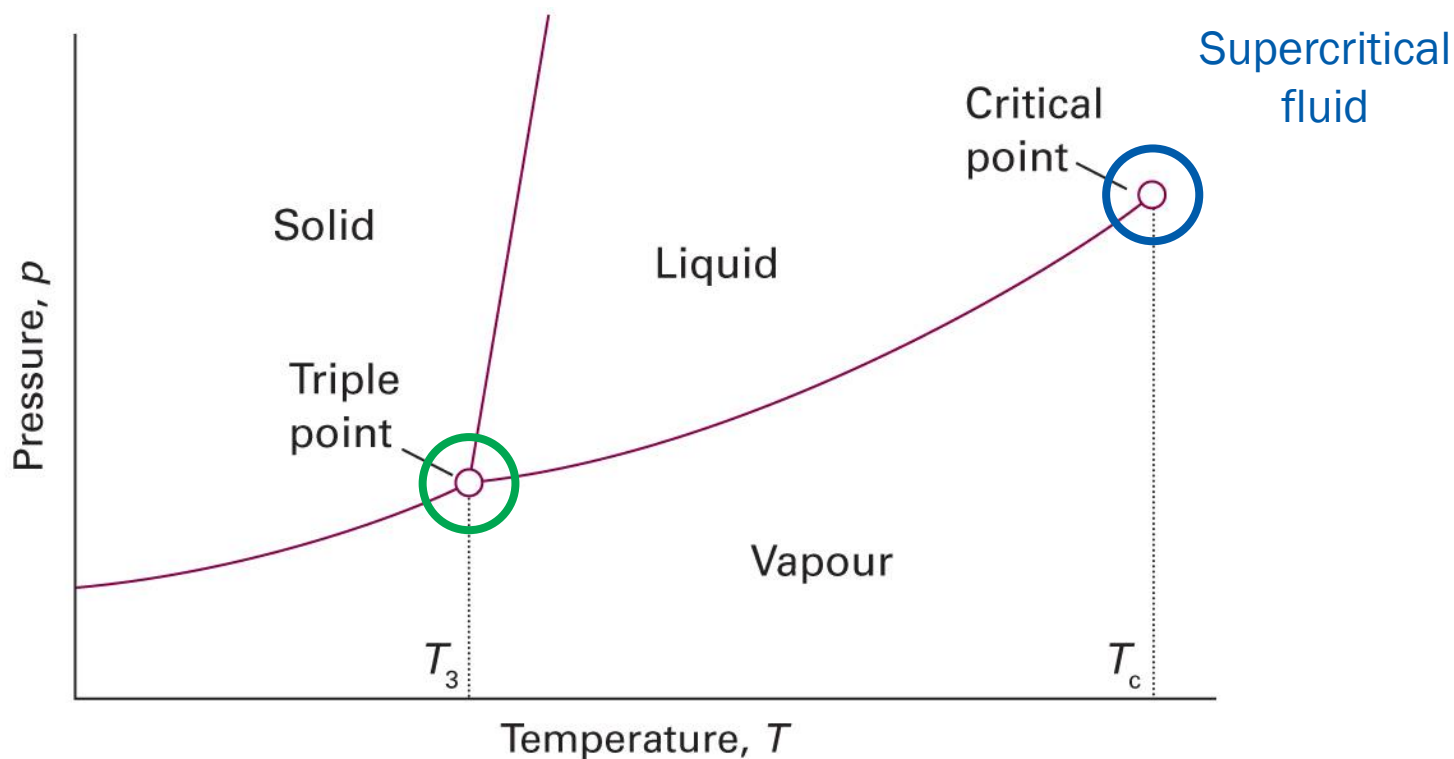
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.4)

Freezing and Boiling Points

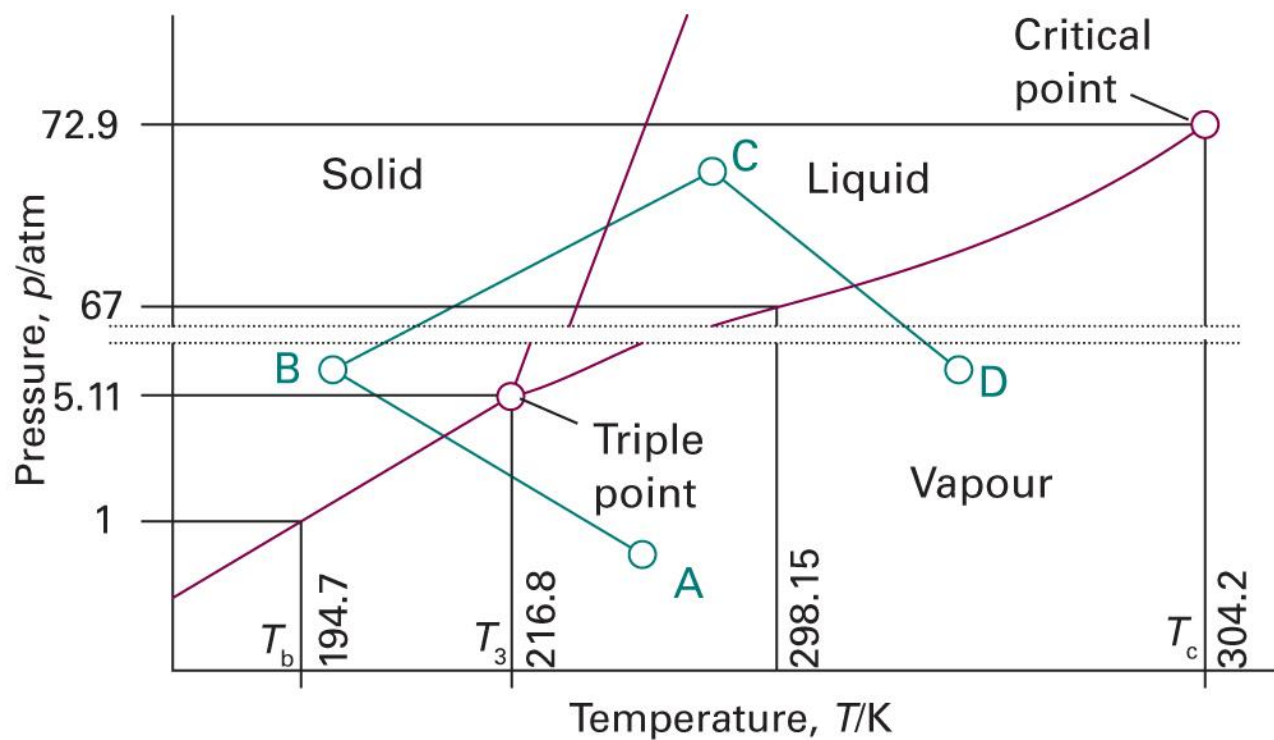
- Normal boiling point T_b : b.p. at 1 atm
- Standard boiling point: b.p. at 1 bar
- e.g. water: $T_b = 100^\circ\text{C}$, standard b.p. = 99.6°C

- Normal freezing point T_f : f.p. at 1 atm
- Standard freezing point: f.p. at 1 bar
- Usually the difference between the two is negligible

Two Characteristic Points

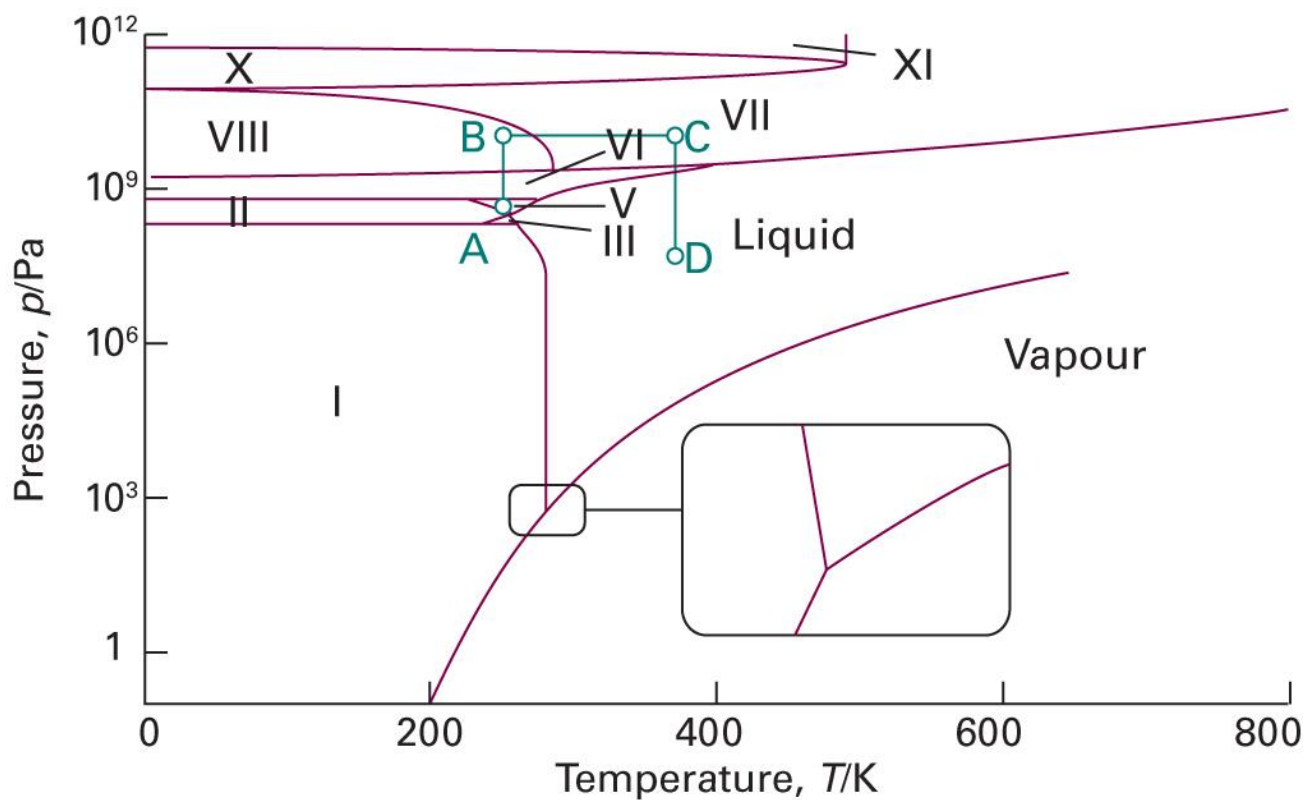


Carbon Dioxide



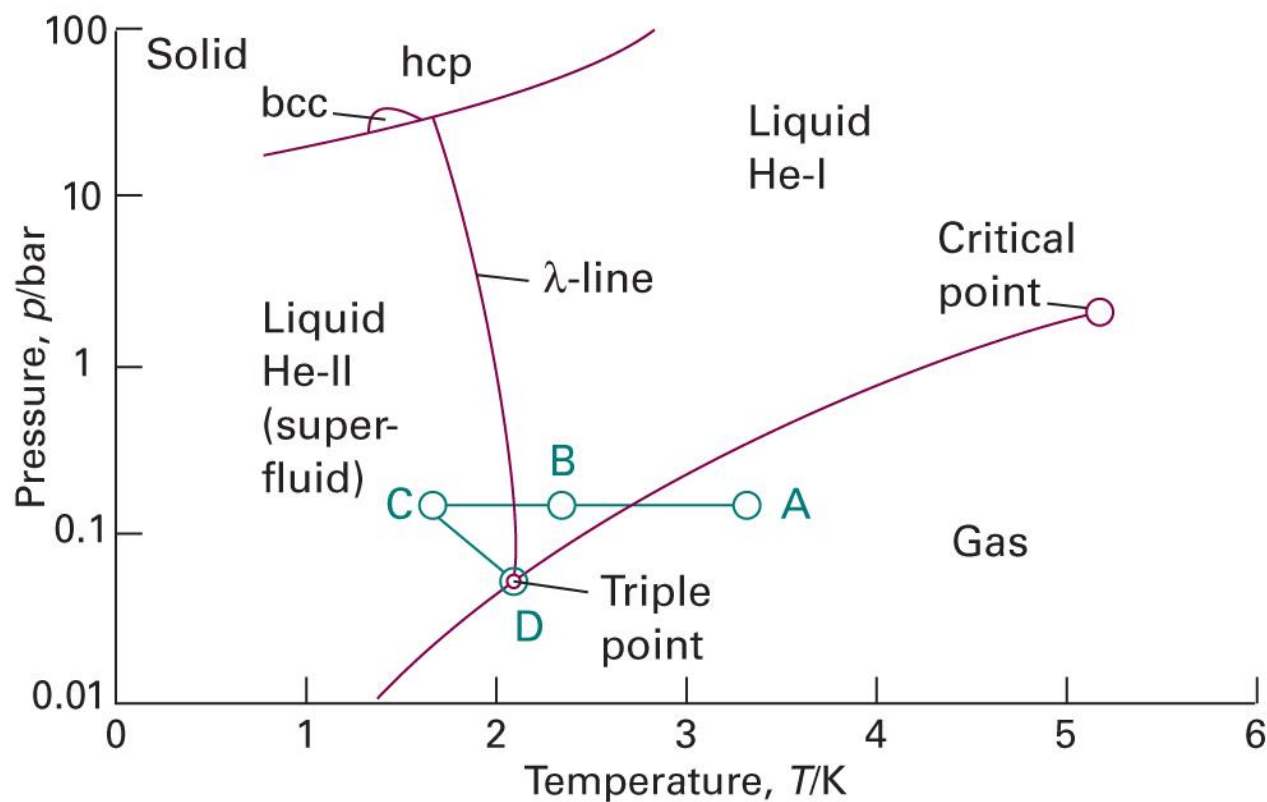
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.8)

Water



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.9)

Helium (^4He)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.11)